NAME :
JATIN
SUNIL SONAWANE

PRN :
202501040405

CS4 DIVISION

EDS Lab Assignment - 2

Loaded the cricket world cup dataset as :

```python
import pandas as pd
import numpy as np

df = pd.read_csv("batting_stats_for_icc_mens_t20_world_cup_2024.csv")
```

1. Display the top 5 batsmen by total runs:

```python
df.nlargest(5, 'Runs')[['Player', 'Team', 'Runs']]
```

	Player	Team	Runs
179	Rahmanullah Gurbaz	Afghanistan	281
202	RG Sharma	India	257
75	TM Head	Australia	255
47	Q de Kock	South Africa	243
82	Ibrahim Zadran	Afghanistan	231

✨ This problem highlights the ***highestrun scorers*** of the tournament.

2. Calculate the average runs scored across all players:

```
df['Runs'].mean()
```

```
np.float64(49.149797570850204)
```

📈 This calculates the ***mean runs scored*** by all players in the dataset.
🌍 It gives a general overview of the tournament’s batting level.
⚖️ Higher averages suggest batting-friendly conditions.
🌧️ Lower averages indicate strong bowling or tough pitches.

🎯 Useful for comparing performance across tournaments.

3. Find which team scored the highest total runs (sum of all its players):

```python
df.groupby('Team')['Runs'].sum().sort_values(ascending=False).head(5)
```

Team	Runs
India	1201
South Africa	1058
West Indies	1056
Australia	981
Afghanistan	922

dtype: int64

👥 This aggregates runs scored by each team collectively.
💪 It reflects the ***overall batting strength*** of a team.
📌 Helps compare teams beyond individual brilliance.
🔥 Teams with higher totals usually have strong

depth.
📊 Useful for understanding team dominance.

4. Find the player with the highest batting average:

```python
df.loc[df['Ave'].idxmax(), ['Player', 'Ave']]
```

	26
Player	RD Berrington
Ave	102.0

dtype: object

📊 This identifies the most ***consistent player*** in the dataset.
💡 Batting average shows reliability and performance stability.
🏏 A higher average means fewer dismissals and steady scoring.
🧠 These players often anchor the innings.
⭐ They are considered dependable assets for the team.

5. List all players with at least one fifty or more

```
fifties = df[(df['50'] >= 1)][['Player', 'Team', '50']]
```

🏏 This filters players who scored ***50+ runs in a match***.
🔥 Half-centuries indicate impactful performances.
📊 Shows players capable of handling pressure situations.
🎯 These innings often influence match outcomes.
💡 Highlights consistent contributors in the team.

6. Find players with a strike rate above 150:

```
high_sr = df[df['SR'] > 150][['Player', 'Team', 'SR', 'Runs']]
```

🚀 Identifies players with ***fast scoring ability***.
📊 Strike rate measures runs scored per 100 balls.
🔥 High SR players are usually aggressive hitters.
💥 Important in T20 for quick run acceleration.
🎯 Often used as finishers in the team lineup.

7. Which team has the highest average strike rate?

```python
df.groupby('Team')['SR'].mean().sort_values(ascending=False).head(5)
```

Team	SR
Scotland	142.087143
West Indies	124.043846
England	120.620000
India	115.641818
Australia	113.418333

📈 Measures overall team ***aggression in batting***.
🔥 High strike rate teams play attacking cricket.
⚔️ Reflects modern T20 strategies.
📊 Helps compare playing styles of teams.
💡 Indicates dominance in scoring pace.

8. Count how many players scored ducks (0 runs):

```
(df['Runs'] == 0).sum()
```

```
np.int64(20)
```

🧠Counts players who scored ***zero runs***.

📉 Indicates failures in batting performance.

⚠️ High duck count may show inconsistency.

🎯 Can reflect strong opposition bowling.

📊 Useful for identifying weak areas

9. Find the player with the highest "Not Outs" (NO):

```python
df.loc[df['NO'].idxmax(), ['Player', 'NO']]
```

	119
Player	KA Maharaj
NO	4

dtype: object

🛡️ Identifies players who remained ***not out frequently***.
🏏 Important for calculating batting average.
🔥 Often finishers who complete matches.
🎯 Shows ability to handle pressure situations.
💡 Indicates reliability in crucial moments..

10. Compute correlation between Runs and Strike Rate:

```
df[['Runs', 'SR']].corr().iloc[0, 1]
```

```
np.float64(0.5755119564198196)
```

📊 Measures relationship between scoring and speed.
🔍 Helps understand batting patterns.
📈 Positive correlation → aggressive & high scoring.
📉 Negative → trade-off between speed & consistency.
💡 Useful for advanced performance analysis.

11. Determine how many unique teams participated:

```python
df['Team'].nunique()
```

```
20
```

📊 Counts total teams present in dataset.
🧠Basic step in data exploration.
✔️ Ensures dataset completeness.
🌐 Shows tournament participation level.
💡 Helps in structuring further analysis.

12. Calculate average runs per team:

```python
df.groupby('Team')['Runs'].mean().sort_values(ascending=False)
```

	Runs
Team	
India	109.181818
South Africa	105.800000
Australia	81.750000
West Indies	81.230769
Scotland	77.714286
England	71.333333
Afghanistan	70.923077
Bangladesh	63.000000
United States of America	59.214286
Canada	40.900000
Netherlands	36.916667
Pakistan	35.384615

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Canada	40.900000
Netherlands	36.916667
Pakistan	35.384615
Sri Lanka	32.000000
Namibia	27.714286
Oman	27.133333
Ireland	26.636364
Papua New Guinea	25.307692
New Zealand	21.928571
Nepal	21.230769
Uganda	11.866667

👥 Calculates average runs foreach team.

⚖️ Helps measure ***team consistency***.

🔥 High average = balanced performance.

📉 Low average = dependency on few players.

💡 Useful for team comparison.

13. Find the player with the highest number of innings (Inns):

```python
df.loc[df['Inns'].idxmax(), ['Player', 'Inns']]
```

	47
Player	Q de Kock
Inns	9

dtype: object

📊 Identifies player with most batting opportunities.
👤 Usually top-order batsmen.
📈 More innings → higher scoring chances.
💡 Helps normalize performance comparison.
🎯 Reflects player selection consistency.

14. Add a new column for “Runs per innings”:

```python
df['Runs_Per_Inns'] = df['Runs'] / df['Inns']
df.head(3)
```

	Player	Team	Mat	Inns	NO	Runs	HS	Ave	SR	100	50	0	Runs_Per_Inns
0	NP Kenjige	United States of America	4	2	0	1	1	0.5	25.00	0	0	1	0.5
1	Aaron Jones	United States of America	6	6	2	162	94*	40.5	135.00	0	1	1	27.0
2	Aasif Sheikh	Nepal	3	3	0	63	42	21.0	88.73	0	0	0	21.0

🧠Creates a new efficiency metric.

📊 Normalizes runs based on innings played.

⚖️ Allows fair comparison between players.

🚀 Higher value = better performance efficiency.

💡 More insightful than total runs alone.

15. Find top 10 consistent scorers (highest average * Runs):

```python
df['Consistency'] = df['Ave'] * df['Runs']
df.nlargest(10, 'Consistency')[['Player', 'Team', 'Consistency']]
```

	Player	Team	Consistency
75	TM Head	Australia	10837.50
34	HC Brook	England	10512.50
26	RD Berrington	Scotland	10404.00
179	Rahmanullah Gurbaz	Afghanistan	9868.72
129	B McMullen	Scotland	9800.00
65	AGS Gous	United States of America	9592.20
202	RG Sharma	India	9434.47
36	JC Buttler	England	9159.20
173	N Pooran	West Indies	8664.00
216	MP Stoinis	Australia	7140.25

📊 Combines ***consistency + productivity***.
🔥 Helps rank players more effectively.
⚖️ Balances reliability and performance.
🏆 High value = top performer.
💡 Useful for advanced player evaluation.

16. Identify players who have both 50+ average and 150+ strike rate:

```
df[(df['Ave'] >= 50) & (df['SR'] >= 150)][['Player', 'Team', 'Ave', 'SR']]
```

	Player	Team	Ave	SR
34	HC Brook	England	72.5	157.60
80	SD Hope	West Indies	53.5	187.71
129	B McMullen	Scotland	70.0	170.73

🔥 Identifies elite-level performers.

⚡ Combines aggression and consistency.

🏏 Rare and valuable combination.

🎯 These players are match-winners.

💡 Highly impactful in T20 cricket.

17. Find out which team's players had the most centuries (100s):

```
df.groupby('Team')['100'].sum().sort_values(ascending=False)
```

Team	100
Afghanistan	0
Australia	0
Bangladesh	0
Canada	0
England	0
India	0
Ireland	0
Namibia	0
Nepal	0
Netherlands	0
New Zealand	0
Oman	0
Pakistan	0
Papua New Guinea	0

Nepal	0
Netherlands	0
New Zealand	0
Oman	0
Pakistan	0
Papua New Guinea	0
Scotland	0
South Africa	0
Sri Lanka	0
Uganda	0
United States of America	0
West Indies	0

🏏 Counts number of 100s per team.

🔥 Centuries indicate dominant performances.

📊 Often lead to match victories.

💪 Shows strength of top-order batsmen.

💡 Highlights big-match players.

18. Get the median strike rate of each team:

```
df.groupby('Team')['SR'].median().sort_values(ascending=False)
```

Team	SR
Scotland	139.720
England	131.775
West Indies	130.550
India	127.610
Australia	122.695
Afghanistan	104.650
Canada	104.620
South Africa	101.015
Pakistan	100.000
United States of America	99.530
Netherlands	98.075
Bangladesh	94.050
Sri Lanka	92.880
Namibia	86.580
New Zealand	84.445
Oman	78.680
Ireland	73.330
Nepal	71.150
Papua New Guinea	64.150
Uganda	43.180

📊 Calculates central tendency of strike rate.
⚖️ Less affected by extreme values.
💡 Provides realistic performance measure.
📈 Helps compare teams fairly.
🎯 Better than mean in skewed data.

19. Find the number of players per team who played at least 3 matches:

```
df[df['Mat'] >= 3].groupby('Team')['Player'].count().sort_values(ascending=False)
```

Team	Player
Afghanistan	12
England	12
West Indies	12
Australia	11
United States of America	11
Bangladesh	11
Papua New Guinea	11
India	11
Uganda	10
Oman	10
Ireland	10
Netherlands	10

Papua New Guinea	11
India	11
Uganda	10
Oman	10
Ireland	10
Netherlands	10
New Zealand	10
Sri Lanka	10
South Africa	10
Nepal	9
Pakistan	9
Namibia	9
Canada	8
Scotland	7

👥 Identifies regularly playing players.

📊 Shows team stability and consistency.

🔄 Reflects selection strategy.

💪 Stable teams perform better overall.

💡 Useful for analyzing team structure.

20.Determine the top 5 partnerships between Avg ×Strike Rate:

```python
df['Performance_Index'] = df['Ave'] * df['SR']
df.nlargest(5, 'Performance_Index')[['Player', 'Team', 'Ave', 'SR', 'Performance_Index']]
```

	Player	Team	Ave	SR	Performance_Index
26	RD Berrington	Scotland	102.0	139.72	14251.440
129	B McMullen	Scotland	70.0	170.73	11951.100
34	HC Brook	England	72.5	157.60	11426.000
80	SD Hope	West Indies	53.5	187.71	10042.485
164	HH Pandya	India	48.0	151.57	7275.360

📊 Combines consistency and scoring speed.
🔥 One of the best T20 performance metrics.
🏆 High index = impactful player.
⚡ Helps rank players efficiently.
💡 Ideal for identifying match-winners.

ALL 20 PROBLEM STATEMENTS ARE MENTIONED NUMBER WISE AND REAL SOLUTIONS ARE CREATED . SCREENSHOTS ARE ADDED FOR EVIDENCE.